

A Novel Fingerprint Database Regeneration Method for Accurate Visible Light Positioning

Shiwu Xu^{ID}, Fen Wei^{ID}, and Yi Wu^{ID}

Abstract—Although the fingerprint-based visible light positioning (VLP) method can achieve centimeter-level accuracy, it requires collecting received signal strength (RSS) values from a large number of training locations in the offline stage, especially when considering the rotation angle of the photodiode (PD). To enhance the practicality of VLP based on the fingerprint method, this article proposes a novel fingerprint database regeneration method. The proposed method does not rely on a signal propagation model but only requires knowledge of the actual coordinates of light-emitting diodes (LEDs) while also considering the impact from the rotation angle of the PD, making it more practical. Simulation and experimental results demonstrate that the proposed method achieves the highest positioning accuracy compared with the other four fingerprint database regeneration methods. The proposed method can still achieve satisfactory localization performance even if the proportion of training locations is low and randomly distributed. When the number of training fingerprints is 60 and randomly distributed, the proposed method can still achieve localization performance similar to that of an actual dense fingerprint database. Even when considering the rotation angle of the PD, the proposed method reduces the average positioning error (PE) by 39.85% in simulation and 29.78% in experiment, respectively, compared with sparse fingerprint databases.

Index Terms—Database regeneration, fingerprint method, receiver rotation, visible light positioning (VLP), weighted K-nearest neighbor (WKNN).

I. INTRODUCTION

THE rapid development of the Internet-of-Things devices, along with their application in indoor scenarios, has made the research and development of indoor positioning systems (IPSs) increasingly important. Although IPSs based on

radio frequency technology, such as Wi-Fi [1], [2], have been developed for a long time, they can only achieve positioning accuracy at the meter level. Recently, researchers have shown increasingly interest in visible light positioning (VLP) methods, because their ability achieves centimeter-level accuracy and utilizes light-emitting diodes (LEDs) directly as signal sources [3]. The commonly used methods for measuring data in VLP systems include received signal strength (RSS), angle of arrival (AOA), and time of arrival (TOA) [3], [4], among which RSS measurements have been typically considered in many VLP works, because the user equipment with the photodiode (PD) can directly measure RSS [5], [6], [7]. VLP methods based on RSS can be classified into ranging and fingerprint methods. The distance between the PD and the LED can be obtained from the Lambertian signal propagation model, and a positioning algorithm is then used to locate the target [8], [9]. Ranging-based VLP methods rely on a highly precise Lambertian signal propagation model. VLP based on fingerprint methods does not rely on the Lambertian signal propagation model and can achieve higher positioning accuracy [10].

Fingerprint-based VLP methods require the location area to be discretized into a large number of grid points, each of which is also called a fingerprint. Target localization can be achieved through the fingerprint matching method due to the uniqueness of coordinates and corresponding RSS feature vectors in different fingerprints. Fingerprint-based VLP methods are divided into two stages: offline and online. During the offline phase, the receiver is positioned at a specific location to collect RSS values from various LED emitters and create an RSS database. During the online stage, the location of receiver is estimated using the RSS fingerprint matching method. Fingerprint-based VLP methods typically employ machine learning algorithms, such as extreme learning machine, multilayer perceptron (MLP), and K-nearest neighbor (KNN) [11], [12], [13], [14], [15]. KNN is a widely used fingerprint localization method due to its high accuracy and low time complexity, making it one of the most popular choices. When KNN fingerprints are assigned different weights, the method is referred to as weighted KNN (WKNN) [10]. In recent years, the MLP has also been successfully applied to VLP based on a fingerprint method [4], [5], [10], [16]. Due to the inherent properties of MLP methods, a large amount of data are needed for learning in the offline phase. MLP methods need to predict output based on what they learn from input data. Therefore, the more the data available for training the model, the better the

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performance of the model. However, collecting a large amount of data can result in high labor and time costs.

The accuracy of positioning using the fingerprint method is directly proportional to the density of fingerprints, meaning that higher fingerprint density results in greater positioning accuracy. However, collecting RSS values from a large number of training locations through offline on-site surveys is not feasible due to the labor and time costs involved, especially when frequent updates of RSS databases are required due to environmental changes. The method of data regeneration can effectively reduce the labor and time costs associated with building an RSS database. A dense RSS database can be generated from a sparse RSS database through a database regeneration method. Wu et al. [17] utilized a linear interpolation (LI) method to reduce the number of training samples required for machine learning. However, to improve the accuracy of regenerated data, it is necessary for the training locations to be evenly distributed. The use of 2-D bivariate spline interpolation technology can achieve high reconstruction accuracy, but the training points should also be evenly and widely distributed [10], [18]. Although using a calibrated propagation model to achieve fingerprint regeneration can result in high positioning accuracy, it requires a precise Lambertian signal propagation model [18], [19], [20], which becomes even more challenging when considering the rotation angle of the PD. Tran and Ha [21] propose an improved cotraining supervised regression (XCoReg) method to estimate unlabeled data, which reduces the burden of data collection. If size of the room and LED lighting parameters are available, fingerprint dataset regeneration can be achieved through DIALux software [22]. When the RSS measurement matrix meets sparsity and orthogonality, the RSS database can be regenerated through compressed sensing (CS) [23], [24], [25]. Although CS methods are widely used in fingerprint positioning based on radio frequency technology, when the density of training locations is low, the accuracy of RSS reconstruction based on CS methods is low due to insufficient measured values. Compared with the existing fingerprint database regeneration methods, the main contributions of this article are as follows.

- 1) A novel method is proposed to regenerate RSS data without relying on a signal propagation model and without requiring a uniform distribution of training fingerprints. In addition, the proposed method takes into account the rotation orientation of the PD and eliminates the need for parallel the irradiance angle of the LED to the incidence angle of the PD.
- 2) To the best of our knowledge, we are the first to investigate the impact from the rotation orientation of the PD on RSS data regeneration in fingerprint-based VLP methods. Thus, the proposed fingerprint database regeneration method is very suitable for practical VLP systems. This is a significant technical improvement over the existing RSS data regeneration in fingerprint-based VLP methods.
- 3) The effectiveness of the proposed method is compared with four other methods. Simulation and experimental results demonstrate that the proposed method can

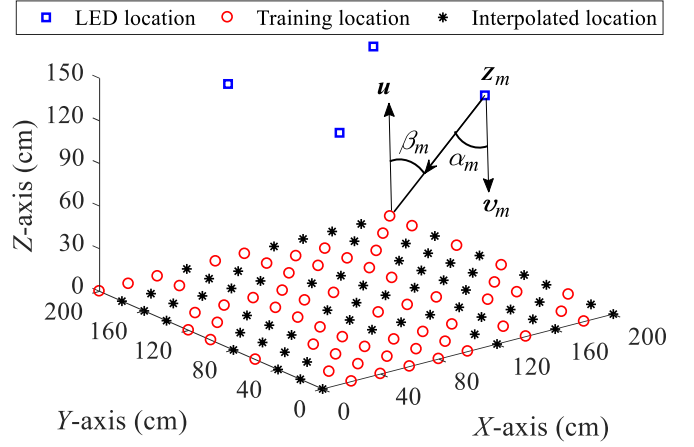


Fig. 1. VLP model based on the database regeneration method.

achieve satisfactory localization performance, even when the proportion of training positions is low and randomly distributed. The research results can provide a new idea for the fingerprint database regeneration method.

The rest of this article is organized as follows. A novel fingerprint database regeneration method is proposed in Section II. Simulation results are conducted in Section III. Experimental results are conducted in Section IV. Finally, the conclusion is presented in Section V.

II. DESIGN OF A NOVEL FINGERPRINT DATABASE REGENERATION METHOD FOR ACCURATE VLP

A VLP model based on the database regeneration method is shown in Fig. 1. The positioning area is divided into a 20×20 cm grid structure, and the coordinates of each grid point and the corresponding RSS vector constitute a fingerprint. To reduce the labor and time costs of on-site investigations, a sparse RSS database can be transformed into a dense one through RSS regeneration methods. First, the RSS values are collected from training locations. Second, based on the RSS matrix of training locations, the RSS matrix at interpolated locations is reconstructed. It should be noted that the method of randomly selecting training locations is adopted to simulate the positioning accuracy of a regenerated RSS database with an uneven distribution of training fingerprints. We assume that the coordinate of the m th LED is $\mathbf{z}_m = (z_{m1}, z_{m2}, z_{m3})^T$, for $m = 1, 2, \dots, M$, where superscript T is the transpose. The numbers of training locations and interpolated locations are N_1 and N_2 , respectively. α_m and β_m are the irradiance and incidence angles, respectively. $\mathbf{v}_m$ represents the orientation vector of the m th LED. $\mathbf{u}$ represents the orientation vector of the PD.

A. RSS Measurement Matrix and Regeneration Matrix

The 3-D coordinates of the i th training location are represented by $\mathbf{x}_i = [x_{1,i}, x_{2,i}, x_{3,i}]^T$, for $i = 1, 2, \dots, N_1$. The matrix composed of coordinate vectors of N_1 training locations is given as follows:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_1}] = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,N_1} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,N_1} \\ x_{3,1} & x_{3,2} & \cdots & x_{3,N_1} \end{bmatrix}. \quad (1)$$

In visible light communication, according to the Lambertian signal propagation model [8], the RSS value received by the PD is anisotropic, meaning that the RSS value received by the PD is affected by its receiving orientation $\mathbf{u}$. We assume that $\varphi_{m,i}^{\mathbf{u}}$ is used to represent the RSS value of the m th LED measured by the PD at the i th training location $\mathbf{x}_i$ with orientation $\mathbf{u}$. In the offline phase, for the given PD receiving orientation $\mathbf{u}$, N_1 training locations in the positioning area are traversed, and the sample data of M LED emission sources are processed, respectively. Then, a complete RSS measurement matrix represented by $\Psi^{\mathbf{u}}$ is constructed, which is given by

$$\Psi^{\mathbf{u}} = [\varphi_1^{\mathbf{u}}, \varphi_2^{\mathbf{u}}, \dots, \varphi_{N_1}^{\mathbf{u}}] = \begin{bmatrix} \varphi_{1,1}^{\mathbf{u}} & \varphi_{1,2}^{\mathbf{u}} & \cdots & \varphi_{1,N_1}^{\mathbf{u}} \\ \varphi_{2,1}^{\mathbf{u}} & \varphi_{2,2}^{\mathbf{u}} & \cdots & \varphi_{2,N_1}^{\mathbf{u}} \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{M,1}^{\mathbf{u}} & \varphi_{M,2}^{\mathbf{u}} & \cdots & \varphi_{M,N_1}^{\mathbf{u}} \end{bmatrix} \quad (2)$$

where $\varphi_i^{\mathbf{u}} = [\varphi_{1,i}^{\mathbf{u}}, \varphi_{2,i}^{\mathbf{u}}, \dots, \varphi_{M,i}^{\mathbf{u}}]^T$, which represents the RSS vector of M LEDs received at the i th training location with a particular PD receiving orientation $\mathbf{u}$. So, the eigenvalue for the i th training location is $(\mathbf{x}_i; \varphi_i^{\mathbf{u}})$, for $i = 1, 2, \dots, N_1$, $\mathbf{u} \in \Omega$, where Ω is the set of PD receiving orientation. In the spherical coordinate system, the value of $\mathbf{u}$ is represented by the polar angle b and azimuth angle r , which will be discussed in detail in Section III.

We assume that the Euclidean distances among N_1 training locations and M LEDs are given by

$$\mathbf{D} = [d_1, d_2, \dots, d_{N_1}] = \begin{bmatrix} d_{1,1} & d_{1,2} & \cdots & d_{1,N_1} \\ d_{2,1} & d_{2,2} & \cdots & d_{2,N_1} \\ \vdots & \vdots & \ddots & \vdots \\ d_{M,1} & d_{M,2} & \cdots & d_{M,N_1} \end{bmatrix} \quad (3)$$

where $\mathbf{d}_i = [d_{1,i}, d_{2,i}, \dots, d_{M,i}]^T$, $d_{m,i} = \|\mathbf{z}_m - \mathbf{x}_i\|_2$, and $\|\cdot\|_2$ denotes the 2-norm on a vector, i.e., the Euclidean distance between two vectors.

The 3-D coordinates of the j th interpolated location are represented by $\mathbf{y}_j = [y_{1,j}, y_{2,j}, y_{3,j}]^T$, for $j = 1, 2, \dots, N_2$. The matrix composed of the coordinate vectors of N_2 interpolated locations is given as follows:

$$\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{N_2}] = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,N_2} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,N_2} \\ y_{3,1} & y_{3,2} & \cdots & y_{3,N_2} \end{bmatrix}. \quad (4)$$

The RSS regeneration matrix corresponding to the coordinates of N_2 interpolated locations is given by

$$\Phi^{\mathbf{u}} = [\phi_1^{\mathbf{u}}, \phi_2^{\mathbf{u}}, \dots, \phi_{N_2}^{\mathbf{u}}] = \begin{bmatrix} \phi_{1,1}^{\mathbf{u}} & \phi_{1,2}^{\mathbf{u}} & \cdots & \phi_{1,N_2}^{\mathbf{u}} \\ \phi_{2,1}^{\mathbf{u}} & \phi_{2,2}^{\mathbf{u}} & \cdots & \phi_{2,N_2}^{\mathbf{u}} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{M,1}^{\mathbf{u}} & \phi_{M,2}^{\mathbf{u}} & \cdots & \phi_{M,N_2}^{\mathbf{u}} \end{bmatrix} \quad (5)$$

where $\phi_j^{\mathbf{u}} = [\phi_{1,j}^{\mathbf{u}}, \phi_{2,j}^{\mathbf{u}}, \dots, \phi_{M,j}^{\mathbf{u}}]^T$ and $\phi_{m,j}^{\mathbf{u}}$ represents the RSS value of the m th LED measured by the PD at the j th interpolated location with orientation $\mathbf{u}$. So, the eigenvalue for

the j th interpolated location is $(\mathbf{y}_j; \phi_j^{\mathbf{u}})$, for $j = 1, 2, \dots, N_2$, $\mathbf{u} \in \Omega$.

We assume that the Euclidean distances among N_2 interpolated locations and M LEDs are given by

$$\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{N_2}] = \begin{bmatrix} e_{1,1} & e_{1,2} & \cdots & e_{1,N_2} \\ e_{2,1} & e_{2,2} & \cdots & e_{2,N_2} \\ \vdots & \vdots & \ddots & \vdots \\ e_{M,1} & e_{M,2} & \cdots & e_{M,N_2} \end{bmatrix} \quad (6)$$

where $\mathbf{e}_j = [e_{1,j}, e_{2,j}, \dots, e_{M,j}]^T$ and $e_{m,j} = \|\mathbf{z}_m - \mathbf{y}_j\|_2$.

Finally, a dense fingerprint database is constructed from fingerprint databases of training points and regeneration points, and only the fingerprint database of training points needs to be conducted offline on-site surveys. This article utilizes the method of fingerprint database regeneration to accurately regenerate RSS values of regenerated points based on those of training points, thereby greatly reducing the labor and time costs of building a dense RSS database.

B. Algorithm Design

The key to the RSS database regeneration method based on nearest neighbor fingerprints is to find L optimal nearest neighbor training fingerprints and regenerate RSS values of the interpolated location through L optimal nearest neighbor training fingerprints. The weight vector $\mathbf{C}$ of L nearest neighbor training fingerprints directly determines the accuracy of interpolated fingerprints. Due to the RSS vector of the interpolated location is unknown, it is evident that we cannot obtain the weight vector $\mathbf{C}$ of L nearest neighbor fingerprints based on the features of RSS. Obtaining an accurate mathematical model for optical signal propagation is challenging due to factors, such as noise interference and changes in receiver orientation. However, as the distance between the LED and the PD increases, the RSS value decreases, which is a fundamental physical law. As both distance matrices, $\mathbf{D}$ and $\mathbf{E}$, are available, the RSS matrix $\Psi^{\mathbf{u}}$ for training fingerprints is known. Obviously, the smaller the Euclidean distance between distance vectors $\mathbf{d}_i$ and $\mathbf{e}_j$, the stronger the correlation between the j th interpolated fingerprint and the i th training fingerprint, which is also a fundamental physical law. This article utilizes these two basic principles, i.e., the correlation between LED-PD distances and RSS values, and the distance correlation between $\mathbf{d}_i$ and $\mathbf{e}_j$, to accurately regenerate RSS values for interpolation locations with a particular PD receiving orientation $\mathbf{u}$. By utilizing two known distance vectors $\mathbf{d}_i$ and $\mathbf{e}_j$, the proposed method can accurately obtain the optimal value L and corresponding weight vector $\mathbf{C}$ for the interpolated fingerprint.

The pseudocode of the proposed method is shown in Algorithm 1. In line 5, $\mathbf{dis} = [\text{dis}_{j,1}, \text{dis}_{j,2}, \dots, \text{dis}_{j,N_1}]^T$, $\text{sort}(\mathbf{dis})$ represents an ascending sorting function, $\mathbf{\Gamma}$ represents the ascending distance vector, and $\mathbf{I}$ represents the corresponding index set. In line 7, $\mathbf{D}_1(:, 1:L)$ represents the L column values of the matrix $\mathbf{D}$ corresponding to the index set $\mathbf{I}$. In line 8, $\mathbf{C} = [C_1, C_2, \dots, C_L]^T$ represents the weights of L nearest neighbor training fingerprints, which was calculated by the linear least squares estimation method.

In line 9, $\tilde{e}_K$ represents the estimated Euclidean distance vector between the j th interpolated location and the M LEDs. In line 13, the equation represents the L value corresponding to the minimum output error E_L .

Algorithm 1 Pseudocode for the Proposed Method

Input: RSS measurement matrix Ψ^u , distance matrix $\mathbf{D}$ and $\mathbf{E}$, the number of maximum nearest neighbor training fingerprints $L_{\max}$.
Output: RSS regeneration matrix Φ^u .

```

1. for  $j = 1 : N_2$ 
2.   for  $i = 1 : N_1$ 
3.      $dis_{ji} = \|\mathbf{e}_j - \mathbf{d}_i\|_2$ 
4.   end for
5.    $[\mathbf{I}, \mathbf{I}] = \text{sort}(dis)$ 
6.   for  $L = 1 : L_{\max}$ 
7.      $\mathbf{D}_1(:, 1:L) = \mathbf{D}(:, \mathbf{I}(1:L))$ 
8.      $\mathbf{C} = (\mathbf{D}_1^T \mathbf{D}_1)^{-1} \mathbf{D}_1^T \mathbf{e}_j$ 
9.      $\tilde{e}_L = \mathbf{D}_1 \mathbf{C}$ 
10.     $E_L = \|\mathbf{e}_j - \tilde{e}_L\|_2$ 
11.     $E(L) = E_L$ 
12.  end for
13.   $L = \arg_L \min(E); \text{ s.t. } 1 \leq L \leq L_{\max}$ 
14.   $\mathbf{D}_1(:, 1:L) = \mathbf{D}(:, \mathbf{I}(1:L))$ 
15.   $\mathbf{C} = (\mathbf{D}_1^T \mathbf{D}_1)^{-1} \mathbf{D}_1^T \mathbf{e}_j$ 
16.   $\phi_j^u = \Psi^u(:, \mathbf{I}(1:L)) \mathbf{C}$ 
17.   $\Phi^u(:, j) = \phi_j^u$ 
18. end for

```

In Algorithm 1, first, from lines 1 to 13, it can be seen that the proposed method can adaptively select the optimal value L according to the minimum matching error E_L between the estimated Euclidean distance vector $\tilde{e}_L$ and the actual Euclidean distance vector $\mathbf{e}_j$. Second, from lines 14 to 15, it can be seen that based on the obtained optimal value L and the corresponding index set $\mathbf{I}$, the optimal nearest neighbor fingerprint matrix $\mathbf{D}_1$ of the j th interpolated location can be obtained. Third, by utilizing two distance matrices $\mathbf{D}_1$ and $\mathbf{e}_j$, the weight vector $\mathbf{C}$ of L nearest neighbor training fingerprints can be calculated by the linear least squares estimation method. Fourth, for a particular orientation $\mathbf{u}$, the RSS matrix Φ^u of interpolated locations is regenerated by the RSS matrix Ψ^u of training locations. Finally, according to the RSS measurement matrix Ψ^u of N_1 training locations and the RSS regeneration matrix Φ^u of N_2 interpolated locations, a dense fingerprint database is obtained.

It can be observed that the proposed method adaptively matches the optimal number of nearest neighbor training fingerprints based on different interpolated fingerprints, thereby improving the accuracy of regenerate RSS data. The proposed method only requires knowledge of the distance between the LED and the interpolated location, without relying on a signal propagation model. In addition, it takes into account the influence from the rotation angle of the PD, making it highly suitable for practical VLP applications.

III. SIMULATION RESULTS

To validate the effectiveness of the proposed method, we considered the following four fingerprint database regeneration methods as the comparison baselines: Lambertian regeneration (LR) method [19], LI method [17], XCoReg method [21], and CS method [23]. In addition, we used actual dense and sparse databases as comparison baselines. As we know, machine learning methods can be used in the online stage to achieve target localization. In [10], [18], and [19], the results demonstrated that the WKNN algorithm achieves high positioning accuracy in VLP based on fingerprint methods. In our previous work [12], the results demonstrated that the WKNN algorithm outperformed other machine learning algorithms in fingerprint-based VLP methods. Due to the widespread use of WKNN in evaluating the positioning accuracy of VLP based on fingerprint database regeneration methods, such as [10], [18], and [19], this article selects WKNN as the reference algorithm to verify the positioning accuracy of the fingerprint database regeneration method, and the distance metric method adopts Euclidean distance. Without loss of generality, this article solely focuses on 2-D positioning. It should be noted that the proposed method can also be extended to 3-D positioning. However, this requires collecting RSS values on more dimensions, which increases the labor and time costs of building an RSS database—a common attribute of fingerprint methods. Fortunately, we can reduce these costs through fingerprint database regeneration methods.

A. Simulation Model

The simulation model employs the widely adopted line-of-sight (LOS) Lambertian signal propagation model [8], [20], [21], [26]. The light power value received by the PD from the m th LED is given as follows:

$$P_r = P_t \frac{A(\lambda + 1)T_s g}{2\pi d_m^2} (\cos(\alpha_m))^\lambda \cos(\beta_m) \quad (7)$$

where P_r and P_t represent the received and transmitted power, respectively, g represents optical concentrator gain, λ is the Lambertian order, $d_m = \|\boldsymbol{\theta} - \mathbf{z}_m\|_2$, and $\boldsymbol{\theta}$ represents the coordinates of the PD. T_s represents optical filter gain. A is physical area of the PD. α_m and β_m are given by [8]

$$\alpha_m = \arccos\left(\frac{(\boldsymbol{\theta} - \mathbf{z}_m)^T \mathbf{v}_m}{\|\boldsymbol{\theta} - \mathbf{z}_m\|_2 \|\mathbf{v}_m\|_2}\right) \quad (8)$$

$$\beta_m = \arccos\left(-\frac{(\boldsymbol{\theta} - \mathbf{z}_m)^T \mathbf{u}}{\|\boldsymbol{\theta} - \mathbf{z}_m\|_2 \|\mathbf{u}\|_2}\right). \quad (9)$$

Without loss of generality, the orientations of LEDs are vertical downward, which is also consistent with the typical LED layout settings. Fig. 2 shows the orientation $\mathbf{u}$ of the PD in the spherical coordinate system, where b is the polar angle between the orientation $\mathbf{u}$ and the positive direction of the Z-axis, and r is the azimuth angle between the orthogonal projection of orientation $\mathbf{u}$ in the XY plane and the positive direction of X-axis, measured counterclockwise.

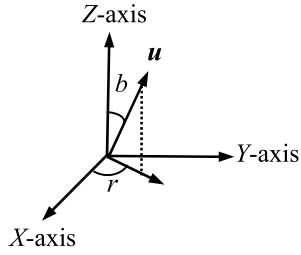


Fig. 2. Orientation $\mathbf{u}$ of the PD in the spherical coordinate system.

Then, the orientation $\mathbf{u}$ of the PD is given by [26]

$$\mathbf{u} = [\sin b \cos r, \sin b \sin r, \cos b]^T. \quad (10)$$

B. Simulation Parameters

The parameters for the LOS Lambertian signal propagation model follow typical LED parameter settings, as follows: $P_t = 6$ W, $T_s = g = 1$, $\lambda = 1$, and $A = 1$ cm². The four LEDs are installed on the ceiling of a 2×2 m² area at a fixed height of 1.5 m from the PD, which is divided into a 20×20 cm grid, i.e., $N_1 + N_2 = 121$. $\mathbf{v}_m = [0, 0, -1]^T$, and $\forall m = 1:4$. The 2-D coordinates of the four LEDs are (0.5, 0.5 m), (1.5, 0.5 m), (0.5, 1.5 m), and (1.5, 1.5 m), respectively. To eliminate the influence of randomness, the following results are an average of 100 executions with the same value of N_1 . The 400 test points appear randomly in a 2×2 m² area. In the WKNN algorithm, $K = 4$, which shows the best positioning performance [10], [18], [19]. In the proposed method, $L_{\max} = 10$. In the XCoReg method, the maximum number of iterations is 50, and the maximum number of nearest neighbors in two consecutive regressors is 5 and 7, respectively, which is based on the corresponding reference. No other control parameters need to be set in LR and CS methods. However, the LR method requires obtaining accurate path loss indices based on the RSS values of training locations. It is worth noting that the LR method in [19] did not consider the impact from the rotation orientation of the PD, while this article considers the impact from the rotation orientation of the PD. Therefore, this article trains the Lambert signal propagation model based on the RSS values collected from the training positions and combines (8) and (9) to obtain the accurate path loss indices of the LR method. Otherwise, when considering the impact from the rotation orientation of the PD, the LR method will cause significant positioning errors (PEs). The simulation and experimental analysis tool is MATLAB R2017a.

We have divided the simulation into two different scenarios, as described below.

- 1) *Scenario 1*: The impact from the rotation angle of the PD is not considered, i.e., the irradiance angle of the LED is parallel to the incidence angle of the PD, which are widely adopted by many VLP papers, such as [18], [19], and [20]. In this scenario, the PD orientation $\mathbf{u}$ of each point of the grid is taken with just one orientation, i.e., the polar angle of N_1 training locations with different space coordinates and N_2 interpolated locations with different space coordinates is equal to 0° .

The polar angle b of test points is also equal to 0° , i.e., $\mathbf{u} = [0, 0, 1]^T$.

- 2) *Scenario 2*: The impact from the rotation angle of the PD is considered, i.e., the irradiance angle of the LED is not parallel to the incidence angle of the PD, which is more in line with the practical VLP system. However, it was not considered in previous VLP works based on fingerprint database regeneration methods. In this scenario, in addition to the special state of $\mathbf{u} = [0, 0, 1]^T$, i.e., $b = 0^\circ$, we also consider the following rotation angles: $b \in \{30^\circ, 60^\circ\}$ and $r \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$, i.e., $\mathbf{u} \in \Omega$, and $\Omega = \{\{b = 0^\circ\}, \{b = 30^\circ, r = 0^\circ\}, \{b = 30^\circ, r = 90^\circ\}, \{b = 30^\circ, r = 180^\circ\}, \{b = 30^\circ, r = 270^\circ\}, \{b = 60^\circ, r = 0^\circ\}, \{b = 60^\circ, r = 90^\circ\}, \{b = 60^\circ, r = 180^\circ\}, \{b = 60^\circ, r = 270^\circ\}\}$. Therefore, based on (10), the PD receiving orientation $\mathbf{u}$ has nine different values at each point of the grid, i.e., there are nine different feature vectors in the orientation $\mathbf{u}$ of N_1 training locations with different space coordinates and N_2 interpolated locations with different space coordinates. The polar angle b of test points is randomly generated between 0° and 60° . The azimuth angle r of test points is randomly generated between 0° and 360° .

Although experimental measurements in [26] demonstrate that the polar angle b is restricted to the range of $[0, \pi/2]$, but the average values of the polar angle b for sitting and walking activities are about 41° and 30° , respectively, with standard deviations less than 9° . Therefore, the maximum value of b in simulation scenario 2 is set to 60° .

C. Simulation Results

When $N_1 = 60$, simulation results of average PE varying with signal-to-noise ratio (SNR) are shown in Fig. 3. As expected, as the SNR increases, the average PE of the simulation results for scenarios 1 and 2 gradually decreases and tends to converge. We can observe that the proposed algorithm demonstrates performance closest to that of the actual dense fingerprint database in terms of average PE. The distribution of training locations is uneven due to the fact that there are only 60 training locations, and random distribution is used. Due to the poor reconstruction accuracy of the LI-based RSS regeneration method, significant PEs can occur even at high SNRs (i.e., from 20 to 30 dB) when training fingerprint distribution is nonuniform and low density. Although the positioning accuracy of the five fingerprint database regeneration methods decreases compared with an actual dense fingerprint database, it significantly reduces the average PE at SNRs above 10 dB when compared with a sparse fingerprint database.

When SNR = 20 dB, simulation results of average PE varying with N_1 are shown in Fig. 4. We can observe that in scenarios 1 and 2, the average PEs of the CS and the LI methods vary significantly with N_1 , especially when N_1 is below 50, both algorithms exhibit relatively large average PEs. The accuracy of RSS database reconstruction for interpolated locations is low in the LI method due to the sparse and nonuniform distribution of training locations. Similarly, the CS method also results in low accuracy in RSS database

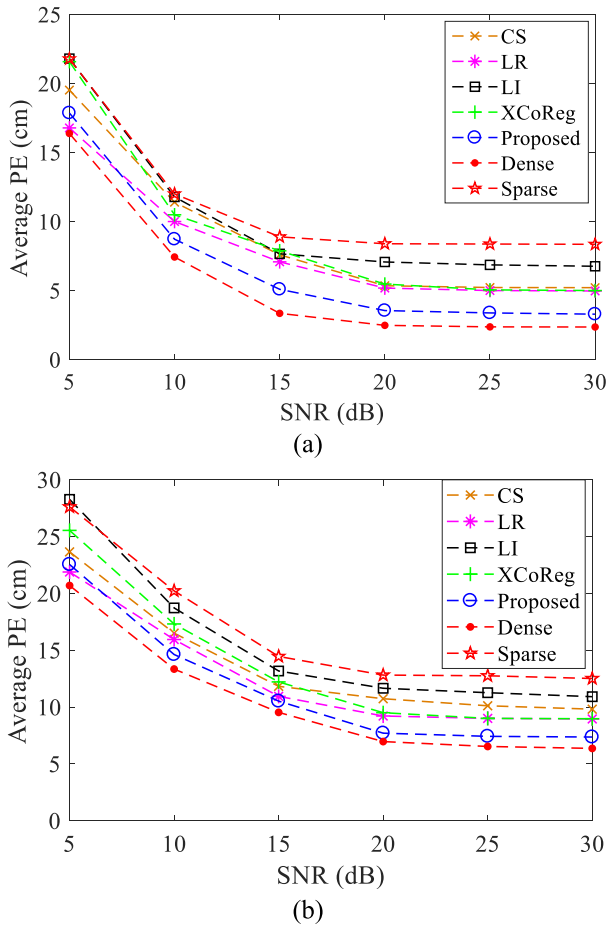


Fig. 3. Simulation results of average PE varying with SNR. (a) Scenario 1. (b) Scenario 2.

reconstruction of interpolated locations due to insufficient measurement values under a low-density training fingerprint distribution. Although the average PE of the LR and proposed methods gradually decreases as N_1 increases, satisfactory positioning results can still be obtained even when N_1 is less than 50, and the average PE is significantly lower than that of a sparse database. The LR method relies on accurate Lambertian signal propagation models and path loss indices, particularly when considering the rotation angle of the PD and different types of noise interference. In such cases, obtaining accurate path loss indices becomes more challenging. However, the proposed method only requires knowledge of the distance between the LED and the interpolated location, making it more practical. We can observe that the average PE of the proposed algorithm exhibits performance closest to that of an actual dense fingerprint database as the N_1 value changes.

When $N_1 = 60$ and SNR = 20 dB, the cumulative distribution functions (cdfs) of PEs are shown in Fig. 5. The cdf can fully describe the probability distribution of a random PE and has been widely used to evaluate the positioning performance of VLP, such as [10], [18], and [19]. We can observe that the proposed method demonstrates performance closest to that of the actual dense fingerprint database in terms of the cdf of PE. In Fig. 5(a), i.e., in scenario 1, the average PEs of the CS, LR, LI, XCoReg, and proposed methods are 5.36, 5.19, 7.08, 5.48, and 3.59 cm, respectively.

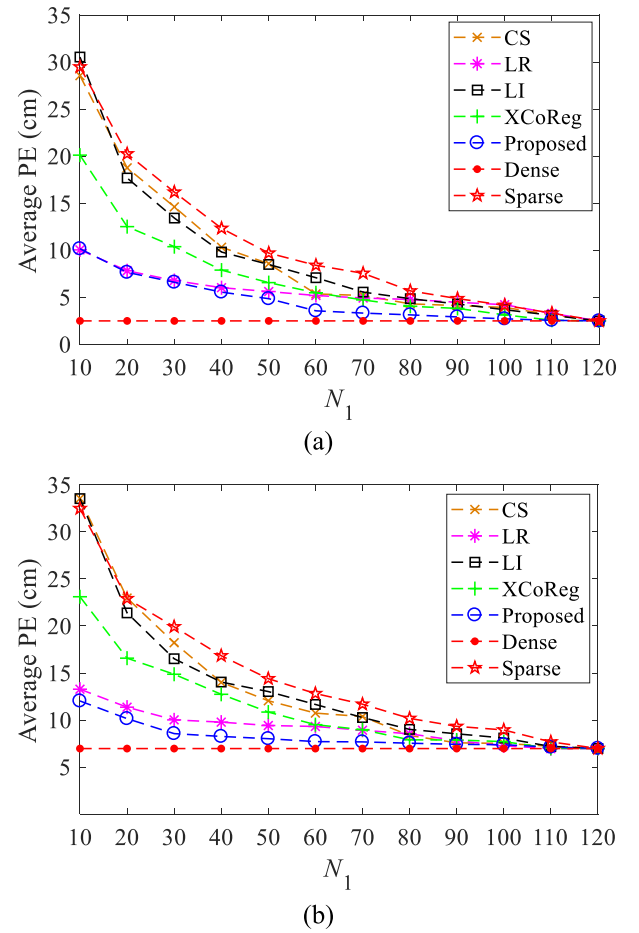


Fig. 4. Simulation results of average PE varying with N_1 . (a) Scenario 1. (b) Scenario 2.

The average PEs in the actual database and sparse database are 2.49 and 8.41 cm, respectively. Compared with the CS, LR, LI, and XCoReg methods, the proposed method has reduced the average PE by 33.02%, 30.82%, 49.29%, and 34.48%, respectively. In addition, compared with the sparse database, the proposed method has reduced the average PE by 57.31%. In Fig. 5(b), i.e., in scenario 2, the average PEs of the CS, LR, LI, XCoReg, and proposed methods are 10.74, 9.22, 11.66, 9.51, and 7.71 cm, respectively. The average PEs in the actual database and sparse database are 6.97 and 12.82 cm, respectively. The proposed method has reduced the average PE by 28.21%, 16.37%, 33.87%, and 18.92%, respectively, compared with the CS, LR, LI, and XCoReg methods. In addition, compared with the sparse database, the proposed method has reduced the average PE by 39.85%.

IV. EXPERIMENTAL RESULTS

Based on our previous VLP work [7], a practical VLP experimental platform based on the fingerprint method is shown in Fig. 6. It mainly consists of four LEDs, a computer numerical control (CNC) machine, and a user equipment with a PD. The 2-D coordinates of four LEDs are (0.5, 0.5 m), (1.5, 0.5 m), (0.5, 1.5 m), and (1.5, 1.5 m), respectively. The vertical distance between LEDs and PD is 1.5 m. The types of PD and LED are silicon p-i-n PD and CREE XML2, respectively. Due to the original light signal collected by the

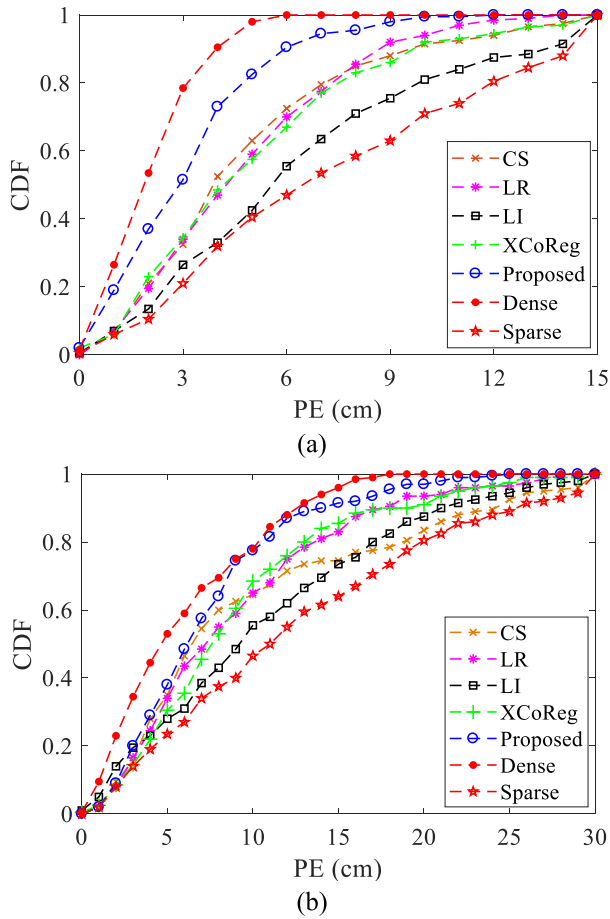


Fig. 5. CDFs of PEs based on the simulation results. (a) Scenario 1. (b) Scenario 2.

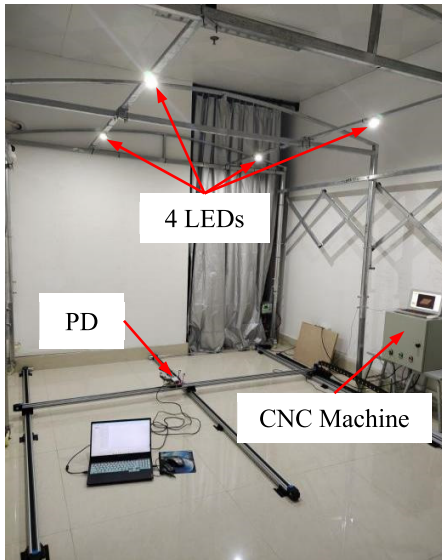


Fig. 6. Practical VLP experimental platform.

PD is weak, this article adopts two-stage op-amp to amplify the original light signal collected by the PD. The schematic of the PD receiving board is mainly composed of a silicon p-i-n PD, two-stage op-amp, a voltage comparator, and a microcontroller unit, which has been widely used in VLP

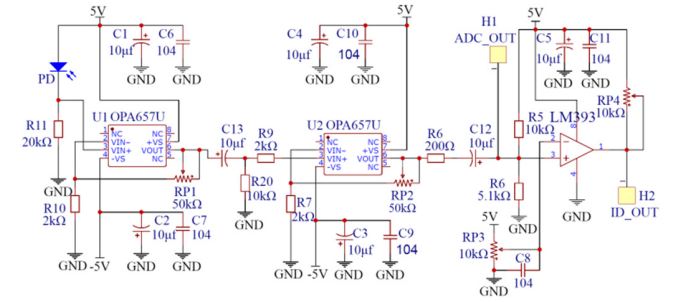


Fig. 7. Schematic of the PD receiving board.

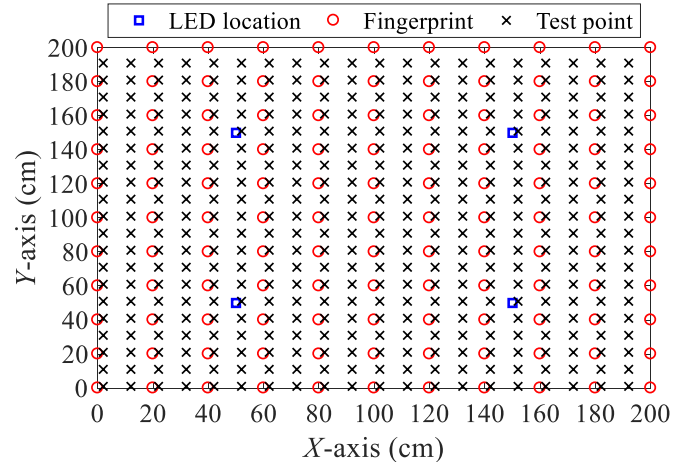


Fig. 8. Actual coordinate distribution of fingerprints and test points.

papers, such as [7], [10], [13], [28], and [29]. Based on [28] and our previous work [7], the schematic of the PD receiving board is shown in Fig. 7. The signal received by the PD passes through two-stage op-amp to generate two different signals. One channel of the signal undergoes analog-to-digital conversion (ADC) to generate the RSS value required for positioning, while the other channel is transmitted to the voltage comparator to convert the signal into a standard high- and low-level signal for the microcontroller unit to recognize the identification (ID) code of the LED. The models of op-amp, voltage comparator, and microcontroller unit are OPA657U, LM393, and STM32F407, respectively. More details can be found in our previous work [7]. The CNC machine was used to measure the actual positions of fingerprints and test points with a measurement accuracy of 0.1 mm, which is significantly better than the expected positioning accuracy of the VLP system. As a result, actual coordinates can be automatically and accurately recorded. The actual coordinates of fingerprints and test points are shown in Fig. 8. The size of the 2-D positioning space is $2 \times 2 \text{ m}^2$, which is divided into a $20 \times 20 \text{ cm}$ grid, i.e., $N_1 + N_2 = 121$. The 400 different test points have been selected to evaluate PEs.

The time-division multiplexing (TDM) technology was used to distinguish the signals from four LEDs. Due to the four LEDs installed on the ceiling being controlled by an STM32F407, TDM is effectively utilized to avoid intercode interference. The transmission scheme at the LEDs is shown in Fig. 9. Each LED is assigned five cycles to send signals.

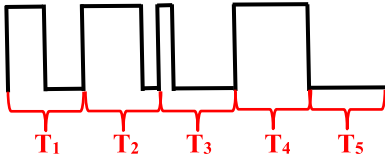


Fig. 9. Transmission scheme at the LEDs.

T_1 cycle represents the starting bit and is represented by a pulsewidth modulation (PWM) wave with a duty cycle of 50%. The ID code is sent in T_2 and T_3 cycles, and the ID codes of four LEDs are represented by codes of 00, 01, 10, and 11, respectively, and 0 and 1 are represented by PWM waves with the duty cycles of 20% and 80%, respectively. As such, the ID code in Fig. 9 is 10. T_4 cycle is represented by a PWM wave with a duty cycle of 100%, which is used to measure the voltage value; then, the RSS is obtained after ADC conversion. T_5 cycle represents the stop bit, which is a low-level signal used to separate the next data frame. More details can be found in our previous work [7]. Of course, it is necessary to explore a more efficient LED transmission scheme to avoid intercode interference, which is beyond the scope of this article, as it aims to solve the regeneration problem of the RSS database.

We have divided the experiment into two different scenarios, as described below.

- 1) *Scenario 3*: The impact from the rotation angle of the PD is not considered. In this scenario, the PD orientation $\mathbf{u}$ of each point of the grid is taken with just one orientation, i.e., the polar angle of N_1 training locations with different space coordinates and N_2 interpolated locations with different space coordinates is equal to 0° . The polar angle b of test points is also equal to 0° , i.e., $\mathbf{u} = [0, 0, 1]^T$.
- 2) *Scenario 4*: The impact from the rotation angle of the PD is considered. In this scenario, in addition to the special state of $\mathbf{u} = [0, 0, 1]^T$, i.e., $b = 0^\circ$, we also consider the rotation angles of fingerprints and test points as follows: $b \in \{30^\circ, 60^\circ\}$ and $r \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$, i.e., $\mathbf{u} \in \Omega$, and $\Omega = \{\{b = 0^\circ\}, \{b = 30^\circ, r = 0^\circ\}, \{b = 30^\circ, r = 90^\circ\}, \{b = 30^\circ, r = 180^\circ\}, \{b = 30^\circ, r = 270^\circ\}, \{b = 60^\circ, r = 0^\circ\}, \{b = 60^\circ, r = 90^\circ\}, \{b = 60^\circ, r = 180^\circ\}, \{b = 60^\circ, r = 270^\circ\}\}$. Therefore, based on (10), the PD receiving orientation $\mathbf{u}$ has nine different values at each point of the grid, i.e., there are nine different feature vectors in the orientation $\mathbf{u}$ of N_1 training locations with different space coordinates and N_2 interpolated locations with different space coordinates.

Based on the experimental measurement results in [26], the same as in simulation scenario 2, the polar angle b is limited to within 60° . It should be noted that in experimental scenario 4, we did not randomly rotate the PD angle of the test points as we did in simulation scenario 2. For simplicity, we kept the polar angle b and azimuth angle r distribution of both test points and fingerprints consistent. That is, each test point has nine different orientations $\mathbf{u}$, just like the fingerprints.

Experimental results of average PE varying with N_1 are shown in Fig. 10. We can observe that as the value of N_1 increases, the average PE of five fingerprint database regeneration methods in scenarios 3 and 4 gradually approaches

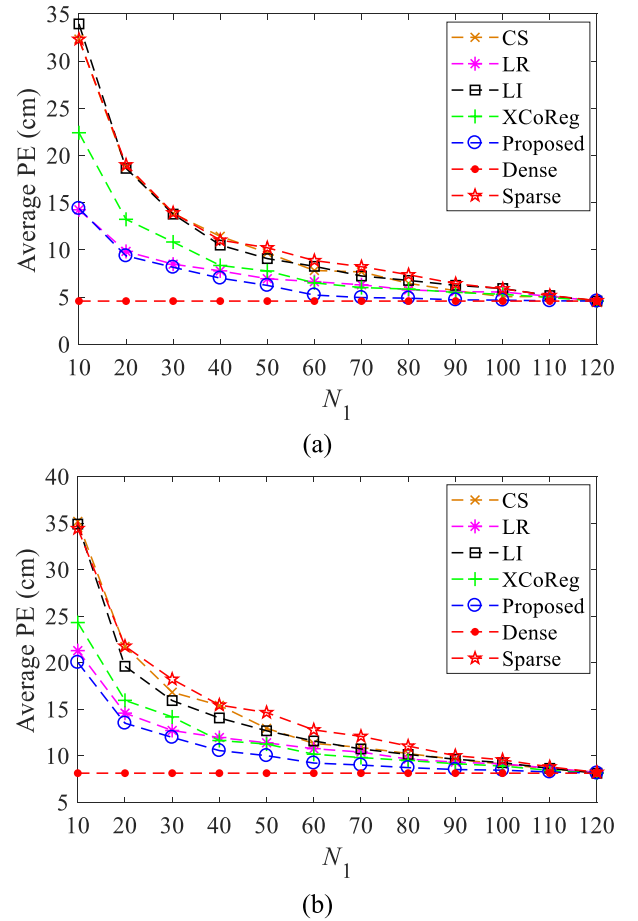


Fig. 10. Experimental results of average PE varying with N_1 . (a) Scenario 3. (b) Scenario 4.

the average PE of the actual dense database. Even when the distribution density of training locations is low and uneven, both LR and proposed methods can still achieve high positioning accuracy in the range of low N_1 values (from 10 to 30), and the average PE is significantly lower than that of a sparse database. The LR method relies on accurate Lambertian signal propagation models and path loss indices, particularly when considering the rotation angle of the PD and different types of noise interference. In such cases, obtaining accurate path loss indices becomes more challenging. However, the proposed method only requires knowledge of the distance between the LED and the interpolated location, making it more practical. The accuracy of RSS regeneration for interpolated locations is not high in the LI method due to the low density and nonuniform distribution of training fingerprints. Similarly, the CS method also leads to low accuracy in RSS reconstruction of interpolated locations due to insufficient measurement values under a low-density training fingerprint distribution. We can observe that the average PE of the proposed method exhibits performance closest to that of an actual dense fingerprint database as the N_1 value changes.

When $N_1 = 60$, the cdfs of PEs are shown in Fig. 11. We can observe that in scenarios 3 and 4, the proposed method demonstrates performance closest to that of the actual dense fingerprint database in terms of the cdf of PE.

In order to further validate the effectiveness of the proposed method, in addition to selecting the WKNN method based on

TABLE I
COMPARATIVE PERFORMANCE IN EXPERIMENTAL SCENARIO 3

	WKNN-ED		WKNN-SCD		MLP		CS	
	APE/cm	P_6	APE/cm	P_6	APE/cm	P_6	APE/cm	P_6
CS	7.82	45.75%	7.24	44.25%	7.11	45.75%	6.84	51.75%
LR	6.63	50.25%	6.32	51.25%	6.30	56.50%	6.18	55.50%
LI	8.26	37.75%	7.62	41.50%	7.34	51.25%	7.10	49.25%
XCoReg	6.51	52.75%	6.23	56.75%	6.11	58.75%	6.69	52.75%
Proposed	5.22	63.50%	4.85	68.75%	4.67	74.75%	4.69	76.25%
Dense	4.58	70.25%	4.15	78.50%	4.51	78.50%	4.18	79.25%
Sparse	8.87	36.75%	8.38	40.25%	8.12	42.75%	7.30	49.25%

TABLE II
COMPARATIVE PERFORMANCE IN EXPERIMENTAL SCENARIO 4

	WKNN-ED		WKNN-SCD		MLP		CS	
	APE/cm	P_{10}	APE/cm	P_{10}	APE/cm	P_{10}	APE/cm	P_{10}
CS	11.33	52.08%	10.34	66.75%	16.26	42.16%	13.26	51.50%
LR	10.76	53.25%	10.04	63.25%	15.52	34.75%	11.73	56.33%
LI	11.59	50.25%	10.60	63.00%	17.40	26.50%	13.19	53.50%
XCoReg	10.47	59.00%	10.12	61.50%	16.49	39.25%	12.14	55.50%
Proposed	9.21	66.75%	7.99	72.50%	13.87	45.50%	10.29	65.25%
Dense	8.13	72.25%	7.78	72.58%	11.57	52.75%	9.61	68.33%
Sparse	12.78	44.00%	11.38	50.08%	20.65	14.08%	17.29	28.75%

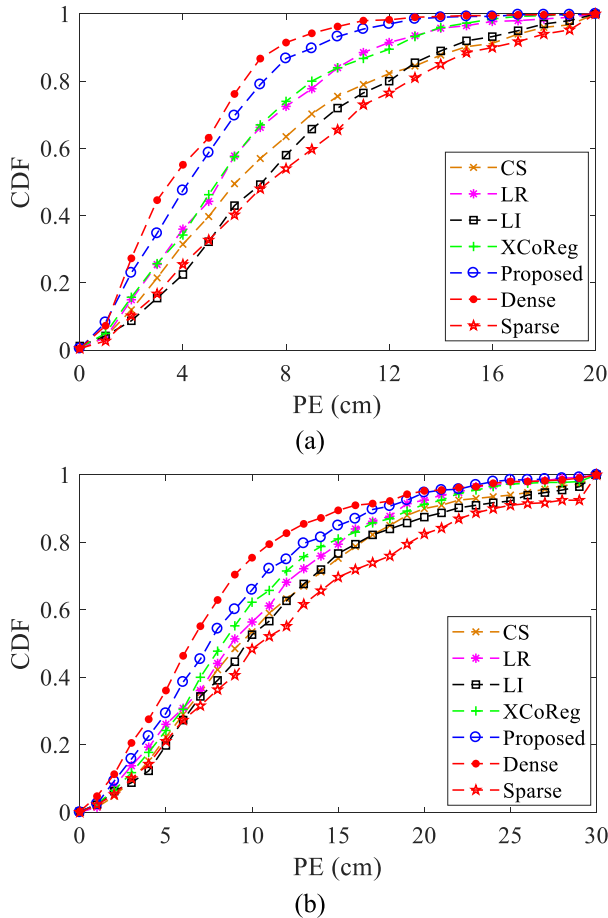


Fig. 11. CDFs of PEs based on the experimental results. (a) Scenario 3. (b) Scenario 4.

Euclidean distance, we also chose the WKNN method based on square chord distance (SCD). The experimental results

in [19] show that SCD is the most robust and accurate metric method compared with other distance metric methods. We also select two positioning methods different from WKNN, i.e., MLP and CS, as reference algorithms to verify the positioning accuracy of the fingerprint database regeneration method. The fingerprint positioning principle of the CS method is the same as in [23]. In order to overcome the susceptibility of the l_1 -norm minimization algorithm to noise, we use the reweighted l_1 -norm minimization (RW l_1 M) algorithm to reconstruct the signal and obtain the weight value of the fingerprint [27]. The sparsity of the CS method is 4, and these nonzero coefficients are normalized to correspond to the weights of the nearest neighbor fingerprints, and the coordinates of the test points are obtained by combining the weighted method. The MLP architecture is the same as in [10], and the hidden layer is a single-layer structure. After comprehensive evaluation of the positioning performance, the optimal number of hidden nodes for experimental scenario 3 and scenario 4 is 15 and 30, respectively. When $N_1 = 60$, the positioning performance comparison results of experimental scenarios 3 and 4 are shown in Tables I and II, respectively. In Tables I and II, WKNN-ED represents the WKNN method based on Euclidean distance. WKNN-SCD represents the WKNN method based on SCD. APE represents average PE. P_6 represents the probability that the PE is less than or equal to 6 cm. P_{10} represents the probability that the PE is less than or equal to 10 cm.

From Table I, it can be seen that the MLP method exhibits the best positioning performance when the rotation angle of the PD is not considered. In Table I, when selecting the MLP method as the reference algorithm to verify the positioning accuracy of the fingerprint database regeneration method, the proposed method has reduced the average PE by 34.31%,

25.87%, 36.37%, and 23.56%, respectively, compared with the CS, LR, LI, and XCoReg methods. Compared with the sparse database, the proposed method has reduced the average PE by 42.48%. In addition, when selecting the MLP method as the reference algorithm, the proposed method can achieve localization performance similar to that of an actual dense fingerprint database.

From Table II, it can be seen that the WKNN-SCD method exhibits the best positioning performance when the rotation angle of the PD is considered. In Table II, when selecting the WKNN-SCD method as the reference algorithm to verify the positioning accuracy of the fingerprint database regeneration method, the proposed method has reduced the average PE by 22.72%, 20.41%, 24.62%, and 21.04%, respectively, compared with CS, LR, LI, and XCoReg methods. Compared with the sparse database, the proposed method has reduced the average PE by 29.78%. In addition, when selecting the WKNN-SCD method as the reference algorithm, the proposed method can achieve localization performance similar to that of an actual dense fingerprint database.

From Tables I and II, we can also observe that whether WKNN, MLP, or CS methods are used to achieve target localization in the online stage, compared with the other four fingerprint database regeneration methods, the average PE of the proposed method exhibits performance closest to that of an actual dense fingerprint database and is significantly lower than that of a sparse database. When the impact from the rotation angle of the PD is not considered, the WKNN, MLP, and CS methods exhibit comparable positioning performance. However, when considering the impact from rotation angle of the PD, MLP requires more RSS data for training due to the RSS database containing more dimensional information. The MLP method needs to predict output based on what they learn from input data. Therefore, the more the data available for training the model, the better the performance of the model. When the required dataset for training is small, it exhibits poor positioning performance. Since the VLP based on fingerprint method has strong sparsity, that is, the target may only appear near one or a small number of fingerprint locations, the positioning method based on CS can also show good positioning performance.

In order to further analyze the positioning performance of the proposed fingerprint database regeneration method under different polar angles, we further added the following two scenarios when the rotation angles of fingerprints and test points are the same.

- 1) *Scenario 4.1*: The impact from the rotation angle of the PD is considered. In this scenario, each point of the grid is taken with just one polar angle and four azimuth angles, but the polar angle and azimuth angle of fingerprints and test points are the same. The rotation angles of fingerprints and test points are as follows: $b = 30^\circ$ and $r \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$.
- 2) *Scenario 4.2*: The impact from the rotation angle of the PD is considered. In this scenario, each point of the grid is taken with just one polar angle and four azimuth angles, but the polar angle and azimuth angle of fingerprints and test points are the same. The rotation

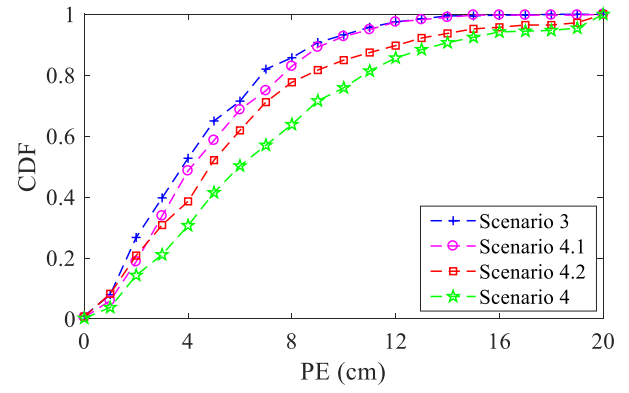


Fig. 12. CDFs of PEs when the rotation angles of fingerprints and test points are the same.

TABLE III
COMPARATIVE PERFORMANCE IN EXPERIMENT
UNDER DIFFERENT SCENARIOS

	Scenario 3	Scenario 4.1	Scenario 4.2	Scenario 4
APE	4.85 cm	5.33 cm	6.32 cm	7.99 cm
P_6	68.75%	64.00%	57.25%	47.25%
P_{10}	92.50%	91.00%	83.50%	72.50%

angles of fingerprints and test points are as follows: $b = 60^\circ$ and $r \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$.

In the online positioning stage, we selected WKNN-SCD as the reference algorithm to verify the positioning accuracy of the proposed fingerprint database regeneration method. When $N_1 = 60$, and the rotation angles of fingerprints and test points are the same, the cdfs of PEs are shown in Fig. 12.

The comparison results of positioning performance under different scenarios are shown in Table III. Fig. 12 and Table III show that as the polar angle increases, the positioning performance deteriorates, especially in the mixed polar angle mode, i.e., experimental scenario 4, where the positioning performance is the worst. In experimental scenario 4, each fingerprint and test point have nine different orientations $\mathbf{u}$, which makes fingerprint matching errors more likely to occur, and thus, the PE becomes larger. The positioning performance will decline with the increase in the polar angles. The main reason is that with the increase in the PD polar angle, the received signal of the PD becomes weaker, and the receiving performance of the PD decreases, resulting in the WKNN-SCD method being unable to accurately match optimal neighbor fingerprints during the positioning process. But, at the same time, it can be seen from Fig. 12 and Table III that the proposed method can still achieve good positioning performance under different polar angles. The accuracy of the fingerprint regeneration method proposed in this article will not be affected by the polar angle of the PD.

The distribution of PEs for 400 test points is shown in Fig. 13. It should be noted that since the PD receiving orientation $\mathbf{u}$ of each test point in scenario 4 is nine different combinations, in order to present the results more clearly, we only list two representative results in scenario 4. Case 1, the rotation angles of test points are as follows: $b = 30^\circ$

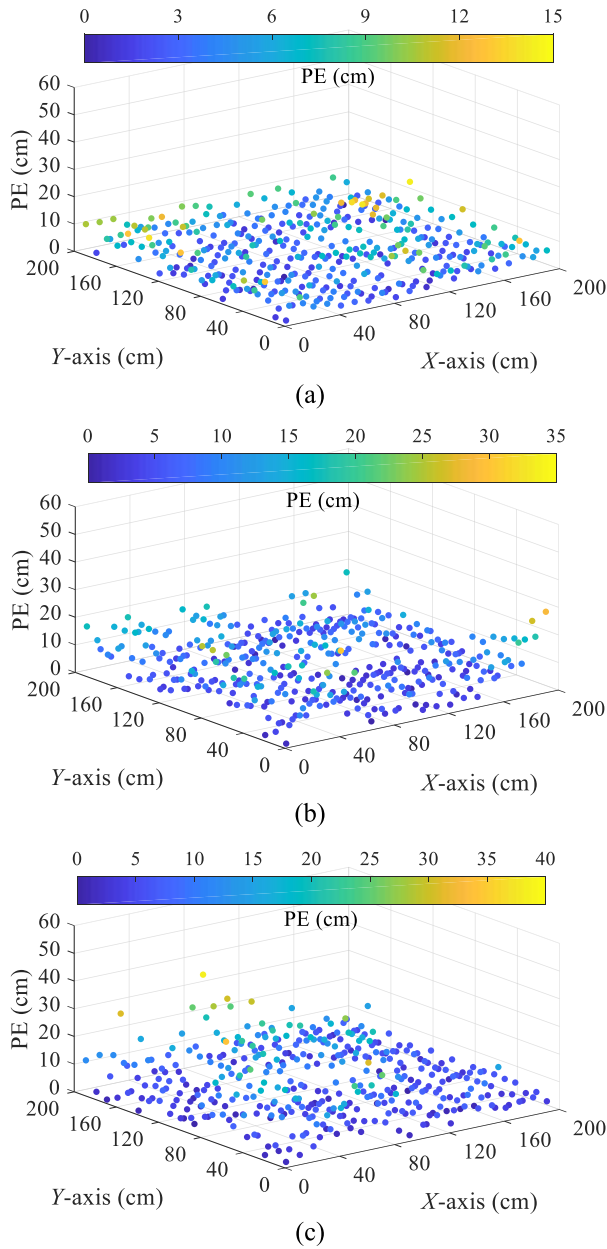


Fig. 13. Distribution of PEs. (a) Scenario 3. (b) Scenario 4, case 1. (c) Scenario 4, case 2.

and $r = 0^\circ$; case 2, the rotation angles of test points are as follows: $b = 60^\circ$ and $r = 0^\circ$. The maximum and average PEs in Fig. 13(a) are 14.48 and 4.85 cm, respectively. The maximum and average PEs in Fig. 13(b) are 30.58 and 7.91 cm, respectively. The maximum and average PEs in Fig. 13(c) are 38.49 and 8.42 cm, respectively. From Fig. 13, it can be seen that test points with significant PEs mainly occur in the boundary area. The main reason is that in the boundary area, the distance between the PD and some LEDs increases, the incidence angle of the PD increases, the received signal of the PD weakens, and the receiving performance of PD decreases. As a result, the WKNN-SCD method cannot match adjacent fingerprints exactly during the positioning process, resulting in large PEs. It can be seen from Fig. 13 that the proposed method can still achieve good positioning performance even when the polar angle is 30° and 60° , respectively. The proposed

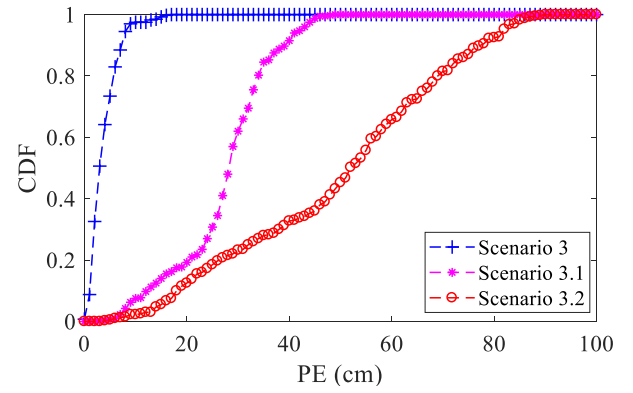


Fig. 14. CDFs of PEs when the rotation angles of fingerprints and test points are different.

method takes into account the rotation orientation of the PD, eliminating the need for parallel the irradiance angle of the LED to the incidence angle of the PD, and is more in line with practical VLP systems, which was not taken into account in previous VLP works based on fingerprint database regeneration methods.

In order to further analyze the impact from the rotation angle of the PD in the process of constructing the RSS measurement matrix, i.e., (2), we added two scenarios with different rotation angles of fingerprints and test points on the basis of experimental scenario 3.

- 1) *Scenario 3.1*: The impact from the rotation angle of the PD is not considered. In this scenario, only one PD orientation $\mathbf{u}$ was taken for each point of the grid, but the orientation of fingerprints and test points is different. The polar angle of fingerprints is equal to 0° , i.e., $\mathbf{u} = [0, 0, 1]^T$. The polar angle of test points is equal to 30° . $r = 0^\circ$.
- 2) *Scenario 3.2*: The impact from the rotation angle of the PD is not considered. In this scenario, only one PD orientation $\mathbf{u}$ was taken for each point of the grid, but the orientation of fingerprints and test points is different. The polar angle of fingerprints is equal to 0° , i.e., $\mathbf{u} = [0, 0, 1]^T$. The polar angle of test points is equal to 60° . $r = 0^\circ$.

In the online positioning stage, we selected WKNN-SCD as the reference algorithm to verify the positioning accuracy in an actual dense fingerprint database. When $N_1 = 121$, and the rotation angles of fingerprints and test points are different, the cdfs of PEs are shown in Fig. 14. Fig. 14 shows that when the rotation angles of fingerprints and test points are different, the PEs of test points will become very large. The average PEs for scenarios 3, 3.1, and 3.2 are 4.15, 27.73, and 50.09 cm, respectively. The larger the polar angle of the test point deviates from the fingerprint, the greater the PE. The setting of experimental scenarios 3.1 and 3.2 is to verify that traditional fingerprint positioning methods only consider the ideal PD receiving orientation, i.e., $\mathbf{u} = [0, 0, 1]^T$. In this case, when the PD receiving orientation of the test point is not the ideal orientation, that is, when the PD receiving orientation is tilted, there may be significant PEs, especially as the polar angle increases, the PE will become larger. The main

reason is that the PD receiving orientation of the fingerprint is inconsistent with the PD receiving orientation of the test point, resulting in the WKNN-SCD method matching the wrong neighbor fingerprints during the positioning process, resulting in large PEs. From (10), it can be seen that under a specific u value, the eigenvalue ($x_i; \varphi_i^u$) for each training location is unique. Compared with the experimental results of scenarios 3.1 and 3.2, it can be seen from scenario 4 that the PE can be effectively reduced by considering the rotation angle of the PD when building the fingerprint database in the offline stage. Therefore, when building a fingerprint database in the offline phase, we should consider the rotation angle of the PD.

V. CONCLUSION

To enhance the practicality of VLP systems based on fingerprint method, this article proposes a novel fingerprint database regeneration method. The proposed RSS regeneration method does not rely on a signal propagation model and does not require a uniform distribution of training fingerprints. The proposed method only requires knowledge of the distance between the LED and the interpolated location, making it more practical. We also take into consideration the impact from the rotation angle of the PD, which is more in line with practical VLP systems and was not taken into account in previous VLP works based on fingerprint database regeneration methods. When the SNR is 20 dB, the number of training fingerprints is 60 and randomly distributed, without considering the influence from the rotation angle of the PD. In simulation and experiment, the average PE of the proposed method is 3.59 and 4.67 cm, respectively. Taking into account the influence from the rotation angle of the PD, the average PEs in simulation and experiment remain at 7.71 and 7.99 cm, respectively. The proposed algorithm can achieve localization performance similar to that of an actual dense fingerprint database. When not considering the rotation angle of the PD, the proposed method reduces the average PE by 57.31% in simulation and 42.48% in experiment, respectively, compared with sparse fingerprint databases. Even when considering the rotation angle of the PD, the proposed method reduces the average PE by 39.85% in simulation and 29.78% in experiment, respectively, compared with sparse fingerprint databases. The research results can provide a new idea for the fingerprint database regeneration method.

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